

Last Change	Review	Approval
X	X	X

Document History

Version	Date	Changes	JIRA-Ticket	Name
V1	2025-05-23	Release of v0.9 of MEVIS MRLab	n/a	jhu
V2	2025-06-24	Changed T2 value and added gradients for additional spin dephasing	n/a	jhu
V3	2025-07-24	Improved graphical visualization, integrated adjustable building blocks, integrated loops	n/a	jhu

1 Introduction and Overview

This document summarizes the delivery of the **MEVIS MRLab** for graphical MR sequence modification and visualization.
The software allows to investigate the effects of different sequences of radiofrequency pulses as well as magnetic gradient shapes, positions and durations on the magnetization and reconstructed image.

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2 Release Summary

Software Component	MEVIS MRLab
Version	https://gitlab.fme.lan/rc_nextgenimaging/mri/simulation/mevismrlab-/tags/v1.0.2
Embedding Product(s)	Not applicable
Purpose of this Release	Provision of the MEVIS MRLab v1.0.2 software under AGPLv3-clause license for research and educational purposes. Not for clinical use.
OS and Compiler Details	OS: Windows 11 Python Version: 3.11
Embedded Libraries	The following libraries are used by the software. Detailed license information is contained in the delivered "third_party_licenses" folder and in the NOTICE file. - MRZero Core (https://github.com/MRsources/MRzero-Core) - PyPulseq (https://github.com/imr-framework/pypulseq) - PyQt (https://www.riverbankcomputing.com/static/Docs/PyQt5/) - PyTorch (https://github.com/pytorch/pytorch) - Numba (https://github.com/numba/numba/tree/main) - mpl-interactions (https://pypi.org/project/mpl-interactions/) - pyvista (https://github.com/pyvista/pyvista) - pyvistaqt (https://github.com/pyvista/pyvistaqt)
Functionality	The software allows to investigate the effects of radiofrequency pulses on magnetization. The user can control the shape as well as the number and distance of radiofrequency pulses, enabling the creation of spin-echoes and stimulated echoes. In addition, the user has the option to control magnetic gradient shapes, positions and durations which affect the image encoding and thus the reconstructed image. MEVIS MRLab v1.0.2 improves v0.9.1 by the following measures: - Improved the graphical spin visualization - Building blocks are now adjustable, allowing to change parameters such as the flip angle of excitation pulses - Integrated loops as a new concept for building repeated elements inside mr sequences
Regulatory Compliance	THIS SOFTWARE IS PROVIDED "AS IS" AND ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED

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Safety Classification	None
Risk Measures	No risk measures defined.
Known Nonconformities and Restrictions	The software is not qualified for use as a medical product or as part thereof. No bugs or restrictions are known.